26. In an organic compound, phosphorus is estimated as

a) $Mg_2P_2O_7$ b) $Mg_3(PO_4)_2$

c) H_3PO_4 d) P_2O_5

27. Ortho and para-nitro phenol can be separated by

a) azeotropic distillation

b) destructive distillation

c) steam distillation

d) cannot be separated

28. The purity of an organic compound is determined by

a) Chromatography

b) Crystallisation

c) melting or boiling point

d) both (a) and (c)

29. A liquid which decomposes at its boiling point can be purified by

a) distillation at atmospheric pressure

b) distillation under reduced pressure

c) fractional distillation

d) steam distillation.

30. Assertion: $CH_3 - \underset{\displaystyle COOC_2H_5}{\underset{|}{C}} = CH - COOH$ is

3– carbethoxy -2- butenoicacid.

Reason: The principal functional group gets lowest number followed by double bond (or) triple bond.

(a) both the assertion and reason are true and the reason is the correct explanation of assertion.

(b) both assertion and reason are true and the reason is not the correct explanation of assertion.

(c) assertion is true but reason is false

(d) both the assertion and reason are false.

II. Write brief answer to the following questions.

31. Give the general characteristics of organic compounds?

32. Describe the classification of organic compounds based on their structure.

33. Write a note on homologous series.

34. What is meant by a functional group? Identify the functional group in the following compounds.

(a) acetaldehyde (b) oxalic acid

(c) di methyl ether (d) methylamine

35. Give the general formula for the following classes of organic compounds

(a) Aliphatic monohydric alcohol

(b) Aliphatic ketones.

(c) Aliphatic amines.

36. Write the molecular formula of the first six members of homologous series of nitro alkanes.

37. Write the molecular and possible structural formula of the first four members of homologous series of carboxylic acids.

38. Give the IUPAC names of the following compounds.

(i) $(CH_3)_2CH–CH_2–CH(CH_3)–CH(CH_3)_2$

(ii) $CH_3\text{-}\underset{\displaystyle CH_3}{\underset{|}{CH}}\text{-}\underset{\displaystyle Br}{\underset{|}{CH}}\text{-}CH_3$

(iii) $CH_3\text{-}O\text{-}CH_3$

(iv) $CH_3\text{-}CH_2\text{-}\underset{\displaystyle OH}{\underset{|}{CH}}\text{-}CHO$

(v) $CH_2 = CH\text{-}CH{=}CH_2$

(vi) $CH_3\text{-}C{\equiv}C\text{-}\underset{\displaystyle Cl}{\underset{|}{CH}}\text{-}CH_3$

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